CAMBRIDGE O LEVEL COMPUTER SCIENCE (2210 / 0478)

Paper 1: Computer Systems | Topic-Wise Question Bank (2024 – 2026 Series)

TOPIC 03: HARDWARE

Exam Sources: Cambridge 2210 / 0478 (May/June 2023, Oct/Nov 2023, May/June 2024)

Question 1 [2210/12/M/J/24 Q5 (2024 Series)] [Total: 7 marks]

(a) State the role of the following CPU registers in the Von Neumann model:
(i) Program Counter (PC)
(ii) Memory Address Register (MAR)
(iii) Memory Data Register (MDR)
(iv) Accumulator (ACC) **[4 marks]**

(b) Describe the role of the Arithmetic Logic Unit (ALU) and Control Unit (CU). **[2 marks]**

(c) Name the three system buses and identify which one is unidirectional. **[1 mark]**

✓ Marking Scheme / Model Answer:

(a) (i) PC: Holds memory address of next instruction to be fetched [1 mark].
(ii) MAR: Holds address in memory being accessed for read/write [1 mark].
(iii) MDR: Stores data/instruction fetched from or waiting to be written to memory [1 mark].
(iv) ACC: Temporarily holds results of arithmetic/logical operations [1 mark].
(b) ALU: Performs math and logic operations [1 mark]. CU: Directs operations and manages control signals across CPU [1 mark].
(c) Address Bus (unidirectional), Data Bus (bidirectional), Control Bus (bidirectional) [1 mark].

Question 2 [2210/12/M/J/23 Q4 (2023 Series)] [Total: 6 marks]

(a) Explain how the following CPU factors affect computer performance:
(i) Clock Speed (measured in GHz)
(ii) Cache Memory size
(iii) Number of CPU Cores **[3 marks]**

(b) Describe what happens during the Fetch stage of the Fetch-Decode-Execute (FDE) cycle with reference to CPU registers. **[3 marks]**

✓ Marking Scheme / Model Answer:

(a) (i) Clock speed: Number of FDE cycles per second; higher clock speed = faster instruction execution [1 mark].
(ii) Cache size: Faster memory located close to CPU; larger cache stores more frequently used instructions, reducing RAM access time [1 mark].
(iii) Cores: Multiple independent processing units; more cores allow parallel execution /

simultaneous processing of multiple tasks [1 mark].
(b) Fetch stage:
1. PC content copied to MAR [1 mark].
2. PC incremented by 1 [1 mark].
3. Instruction in memory at address in MAR copied via Data Bus into MDR, then to CIR [1 mark].

Question 3 [2210/12/O/N/23 Q3 (2023 Series)] [Total: 5 marks]

(a) Explain what is meant by Virtual Memory and why it is needed when physical RAM is full. **[3 marks]**

(b) State two advantages of Solid State Drives (SSD) compared to Magnetic Hard Disk Drives (HDD). **[2 marks]**

✓ Marking Scheme / Model Answer:

(a) Virtual memory is a section of secondary storage (HDD/SSD) used as an extension of RAM when physical RAM is full [1.5 marks].
Inactive pages of data are moved from RAM to secondary storage (swap space) to free up RAM for active processes [1.5 marks].
(b) SSD benefits: [Any 2]
- No moving parts (more durable / shock resistant)
- Much faster read/write access speeds (faster boot and load times)
- Lower power consumption / runs cooler and completely silent

---